The Post-API Age

Week 3 · Foundations module · Textbook Chapter 3

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This week at a glance

The web used to invite us in. This week we ask *why* that changed — and how to keep doing public-interest research anyway.

Monday | Concepts & notebook intro

- The golden age, then enclosure, exemption & erosion

Wednesday | Notebook lab

- A discussion notebook: build a timeline, apply the framework, plan for closure

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 3

Companion notebook

`ch-03-post-api.ipynb`

A conceptual week

No code today — the notebook is a space for notes, arguments, and evidence. Bring a laptop and an example that matters to you.

1. Monday · Concepts

The golden age of web data access

For roughly a decade — about **2008 to 2018** — the web behaved as if it wanted to be studied.

- Twitter opened its API in 2006; Facebook launched the Graph API in 2010; Reddit's API was free and essentially unrestricted; Google exposed trends, maps, and books.
- This was not charity: platforms traded **data access** for **ecosystems**, credibility, and lock-in.
- The bargain built a field — computational social science (Lazer et al. 2009) — and a generation of data journalism and civic tech.

When researchers today say the **"post-API age"** (Freelon 2018), this is the era they measure against.

Built on the bargain

The Wikipedia and Reddit studies that trained a generation — including the instructor's — ran on open APIs, public dumps, and volunteer archives like Pushshift (Baumgartner et al. 2020). No permission, partnership, or payment required.

You will use those same open interfaces in Ch. 10 (Wikipedia).

Textbook Ch. 3, "The Golden Age of Web Data Access."

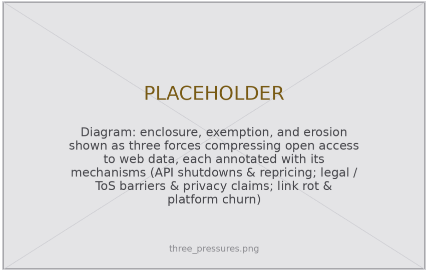
Pressure 1 — Enclosure

Three pressures explain the shift since ~2018: enclosure, exemption, erosion.

Enclosure privatizes a shared resource — the name echoes the fencing of English common land. The posts still exist; the *terms of access* are fenced.

- **Twitter/X**: a free academic archive (2021) → discontinued; enterprise access from $42,000/month. Hundreds of tools went dark.
- **Reddit**: free since 2008 → $0.24 per 1,000 calls (2023); Apollo shut down; 7,000+ subreddits went dark; Pushshift was cut off.

The driver: generative AI. Once text became training data, an open API stopped looking like ecosystem-building and started looking like giving away inventory (Reddit–Google: ~$60M/year).



Three forces fencing the open web.

Textbook Ch. 3, “Enclosure.” Facebook’s CrowdTangle shutdown (2024) was the template.

Pressure 2 — Exemption

Exemption: platforms operate beyond outside scrutiny while running the same studies internally at will.

The asymmetry is the tell. Every A/B test is a behavioral study, and internal teams have total access — but outside researchers meet:

- cease-and-desist letters citing the Computer Fraud and Abuse Act;
- Terms of Service that prohibit the very collection research needs (Fiesler et al. 2020);
- account bans — e.g., Meta disabling NYU's Ad Observatory (2021), justified as *protecting privacy*.

When "privacy" restricts only accountability research — never ad targeting on the same users — it is functioning as exemption, not protection.

Consenting users' data, still shut down.

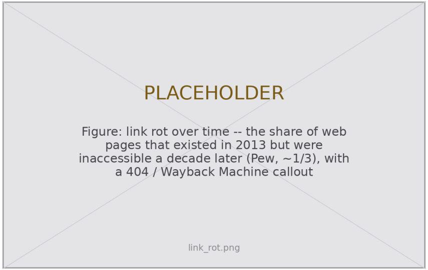
Textbook Ch. 3, "Exemption." Courts push back at the margins (*Sandvig*, *Van Buren*, *hiQ*; see Ch. 2). EU DSA Article 40 may mandate vetted-researcher access.

Pressure 3 — Erosion

Erosion is the quietest pressure: enclosure and exemption are things platforms *do*; erosion is what happens to the web *itself*.

- **Links rot.** About half the links in U.S. Supreme Court opinions are dead; a 2024 Pew study found about a third of 2013 web pages gone a decade later.
- **Platforms churn.** GeoCities vanished; Parler went offline overnight; Twitter became X and broke a decade of embedded links.
- **Sites redesign** and silently break every scraper — fragility you will feel in Ch. 6 & 8.

For research this is a **reproducibility crisis:** a study on the Twitter Academic API cannot be rerun at any price (Davidson et al. 2023).



The record decays unless someone saves it.

Textbook Ch. 3, "Erosion." Two later chapters are counter-erosion: archives (Ch. 7) and automation (Ch. 14).

This is an old problem — public-interest lineages

Data science is not the first profession to face powerful actors gatekeeping information the public needs. Each precedent offers a hard-won lesson:

- **Journalism** — freedom-of-information laws and the muckraking tradition established that records, public *and* private, are legitimate targets of scrutiny. (You work in this lineage when you file a records request.)
- **Law** — discovery rules compel disclosure; due process demands decisions be explainable and reviewable — inherited almost verbatim by algorithmic accountability.
- **Accounting** — after 1929, mandatory audits turned private finances into public record. Lesson: trustworthy disclosure at scale needs **standards and independent examiners**, not goodwill.
- **Urban planning** — impact statements and public comment periods make community input a precondition for decisions about shared resources.
- **Engineering** — licensure and safety codes established duties to the public that override duties to an employer. Data science is still writing its equivalent.

Textbook Ch. 3, "Public Interest Lineages." Missing Manual Ch. 8: "Artifacts Have Politics."

2. Wednesday · Notebook Lab

Working inside it — openness, oversight, ownership

The counter-values are not a fix for the post-API age; they are a **stance** for working inside it. They frame everything we do today.

Openness

Make your *own* work resist enclosure: transparent pipelines, documented collection, and redundant deposits (Zenodo, Internet Archive, open licenses). Wikipedia is the standing counter-example.

Oversight

Insist consequential systems be inspectable by parties who do *not* own them: audits, documentation standards, DSA Article 40. **The skills this course teaches are oversight skills.**

Ownership

Prefer community-governed infrastructure: federated protocols (ActivityPub, AT Protocol), data cooperatives, ArchiveTeam. The long-horizon answer to enclosure.

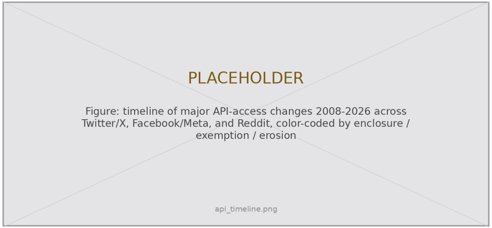
Textbook Ch. 3, "Three Counter-Values." Federated platforms return in Ch. 12 (Social).

Lab 1 — build the timeline

Exercise 1. In the notebook, build a timeline of major API-access changes from **2008 to today**.

- At least **five events** across at least **three platforms**.
- For each, tag which pressure best fits: enclosure, exemption, or erosion.

The point is not memorizing dates — it is seeing the **trend**: free-and-open access for outside observers is what is disappearing, replaced by paid tiers, gated programs, and licensing deals.



Start here; add the events you already know.

Textbook Ch. 3, Exercise 1.

Lab 2 — apply the framework, then plan for closure

Exercise 2 — name the pressure. Pick a platform change *not* in the chapter and write 500 words applying the framework. Ask:

- Was a shared resource fenced? (enclosure)
- Was scrutiny blocked while internal study continued? (exemption)
- Did the record simply decay? (erosion)

Exercise 3 — a contingency plan. Design a project on one web source, then plan for it vanishing mid-study. Redundancy is the strategy; keep ≥ 3 fallbacks in your repertoire:

- **Official research programs** (Meta Content Library, TikTok, Reddit) — real but rationed.
- **Data donation** — users' export rights, with consent built in.
- **FOIA / records requests** — see Ch. 11 (Government).
- **Crowdsourcing** — when no one holds the data, communities create it.

Textbook Ch. 3, Exercises 2–3 & "What This Means for Practice."

Openness in practice — document the collection

The chapter has no scraping code — but openness *does* have a concrete artifact. Record not just **what** you collected but **how**: endpoints, parameters, dates, and the platform's constraints (Gebru et al. 2021). A future reader (or you, in a year) can then reconstruct exactly what was allowed:

```
collection = {
    "source": "reddit.com (PRAW, r/ClimateChange)",
    "endpoint": "GET /r/{sub}/top params: t=year, limit=1000",
    "collected": "2026-02-14",
    "access_tier": "researcher program (application-gated)",
    "constraints": "no redistribution of raw comments; 100 req/min",
    "known_gaps": "pre-2023 history unavailable -- Pushshift offline",
}
import json, pathlib
pathlib.Path("datasheet.json").write_text(json.dumps(collection, indent=2))
# The datasheet travels next to the data; the source may change, the record does not.
```

Textbook Ch. 3, “Openness” — documentation as the practice that survives enclosure.

Lab: work these, then show-and-tell

Work through the notebook prompts in pairs:

1. Timeline: five events, three platforms, tag each pressure
2. Framework application — 500 words on a fresh case
3. Contingency plan — three concrete fallbacks
4. Public-interest audit: take an API-enabled project (paper, investigation, or civic tool) and assess how today's restrictions would hit it. Which value does it embody?
5. 5617: read Davidson et al. (2023) & Perriam et al. (2020); write a 2–3 page "state of research access" brief for one platform, ending with a recommendation.

Show-and-Tell

Come ready to share:

- an access change that surprised you
- a dataset you fear losing
- a provocation for a research design

“If the API closes, I’ll just scrape.”

Visibility, permission, and ethics are three different questions.

Scraping is one tool in a portfolio —
not an escape hatch that makes enclosure irrelevant.

3. Friday · Textbook Revisions

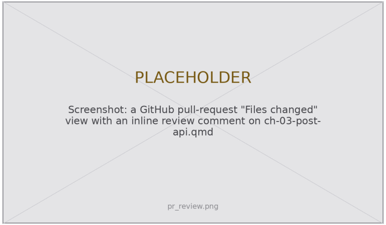
Improving Chapter 3 together

Today we make the book better. Each of you opens a **pull request** against `ch-03-post-api.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

This is a **fast-moving** chapter — prices, lawsuits, and regulations change monthly — so it always needs hands.



Contributions & reviews count toward the Textbook Revisions grade (30%).

A revision menu for Chapter 3

High-value targets — pick one and scope it to a single PR:

- **Add the missing figures.** This chapter is *all prose*. Contribute a three-pressures diagram, an API-access timeline (2008–2026), or a link-rot chart — the deck's placeholders show the need.
- **Update the moving numbers.** Verify and re-cite the Twitter/X pricing, the Reddit–Google deal, and DSA Article 40's implementation status "as of mid-2026." Flag what has gone stale.
- **Resolve the editorial TODOs.** The source carries author-voice and "add citation when published" notes (the openness/oversight/ownership framing). Track down a source or draft the text.
- **Strengthen the open-ended exercises.** Add a rubric, a deliverable format, or a starter table for the timeline so Exercise 1 is self-checking.
- **Extend a Misconception or Further Reading.** Add a post-2023 source on AI-training licensing, tied to "enclosure logic is spreading."
- **Verify the cross-references.** Ch. 3 leans on many `@sec-` links (ethics, wikipedia, archives, automation, social, government). Confirm each resolves and is described accurately.

Targets drawn from the chapter's callouts, thin exercises, "Common Misconceptions," and "Further Reading."

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, source)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Data formats** — XML & JSON, the shapes web data actually arrives in.

Key takeaways

1. The open-API era (~2008–2018) was a **strategic bargain**, not a natural state — and computational social science was built on the trade.
2. **Enclosure** converts collective data into private assets: shutdowns, repricing, and AI-training licensing deals.
3. **Exemption** is the asymmetry of scrutiny — platforms study users freely while blocking those who study platforms.
4. **Erosion** is infrastructural decay, so reproducibility is a problem you must solve *at collection time*.
5. The response is **redundancy**: openness, oversight, ownership — and mastery of many access techniques so no single closure ends your research.

Reading: Textbook Ch. 3. See Freelon (2018); Bruns (2019); Lazer et al. (2020).